

Conclusion

ITSM Service Desk & Ticket Operations Dataset

This project helped me understand how IT Service Management (ITSM) Service Desk & Ticket Operations data can be used for analysis and business understanding. I worked with PostgreSQL for the database part and Python, Pandas, and Matplotlib for data preparation and analysis.

The ticket data and SLA data were brought together in Jupyter Notebook using Pandas. After merging the data, the final dataset contained **360,000 rows and 29 columns**. I then performed data quality checks and used univariate, bivariate, and multivariate analysis to understand the data.

Main Finding

The biggest issue found in the analysis was related to **SLA performance**. Out of 360,000 SLA records, **310,006 were breached**, while **49,994 were met**.

The **Resolution SLA** needs the most attention because all **180,000 Resolution SLA records were marked as breached**. The average actual time for these records was around **5,893 minutes**, which shows a significant difference between the expected and actual resolution time.

The **First Response SLA** performed comparatively better, but it also had a considerable number of breaches. Therefore, both first response time and resolution time should be monitored.

Customer Satisfaction

Customer satisfaction did not show a major difference between breached and met SLA records. The average rating was around **3.01** for both groups.

Because of this, I cannot conclude from this dataset that SLA status alone has a strong effect on customer satisfaction.

Data Quality

During data preparation, there were **no duplicate rows** and no invalid customer satisfaction ratings. The IQR check also did not identify numerical outliers.

There were **61,446 missing customer satisfaction ratings**. These were kept as missing because a missing rating can simply mean that the customer did not provide feedback.

One limitation of the dataset is that several categorical columns had only one value. Because of this, some comparisons did not provide meaningful patterns.

What I Learned

As a fresher, this project helped me understand that data analysis is not only about creating graphs. It is also important to first check the quality and variation of the data and then decide whether a particular analysis actually makes sense.

I learned how to work with database data, merge DataFrames using Pandas, check missing and duplicate values, create derived columns, and perform different types of analysis.

Final Takeaway

Overall, this project gave me practical experience with a complete data analysis workflow. The main area identified for improvement is **SLA performance, especially Resolution SLA performance**.

The project also taught me an important lesson: **business conclusions should be based on what the actual data supports**, while keeping the limitations of the dataset in mind.

This keeps the conclusion **simple, realistic, and appropriate for a fresher portfolio** rather than making it sound like an experienced industry analyst.

THANKYOU!!!!
